H. Search, Discovery & Indexing

Standalone capability brief · Suggested module: **Search SDK + Indexing Platform**

Purpose

Offer reusable indexing and search mechanics while allowing each domain to retain ownership of its business semantics, ranking, and permission model.

1. Scope and capability areas

Search interfaces

Provide consistent full-text and optional semantic search interfaces with filters, facets, pagination, and suggestions.

Index lifecycle

Manage index creation, versioning, reindexing, aliases, and retirement.

Permissions and tenancy

Apply tenant filters and authorization-aware document visibility consistently.

Synchronization

Propagate source changes to indexes with explicit eventual-consistency expectations.

Relevance and analytics

Measure zero-result searches, click-through, latency, and relevance changes without centralizing domain-specific ranking decisions.

2. Shared contracts and standards

- Every indexed document has a source-of-truth identifier, tenant scope, schema version, and update/delete semantics.
- Search results must not reveal documents the caller is not authorized to access.
- Each domain defines searchable fields and business relevance; shared infrastructure provides mechanics.
- Index freshness and consistency guarantees are documented.

3. Recommended implementation approach

- Use a transactional outbox or reliable change feed to reduce lost updates between source data and indexes.
- Support blue-green or alias-based reindexing to avoid partial index replacement.
- Provide standard query limits, pagination tokens, timeouts, and cancellation.
- Run periodic reconciliation between source records and indexed records.
- Test tenant isolation, deletion propagation, and permission changes.

4. Failure behavior and operational expectations

- If the index is stale or unavailable, expose a clear degraded state and avoid returning unauthorized cached results.
- Reindexing should be restartable and should not publish a partially built index as current.

- Deletion and permission revocation should have defined propagation objectives.

5. Suggested operational metrics

- Search latency and error rate
- Index freshness lag
- Zero-result rate
- Reconciliation drift
- Unauthorized-result test failures

6. Readiness and acceptance checklist

- ■ Domains can index and query data through a standard interface.
- ■ Tenant and document permissions are enforced on every query path.
- ■ Reindexing is recoverable and supports safe cutover.
- ■ Freshness and deletion propagation objectives are measurable.

Ownership note: assign a named team, service owner, support route, versioning policy, and deprecation process before treating this capability as a platform contract.